

BUFFALO, SD, USA

WMO: 726627

Lat: 45.604N

Lon: 103.546W

Elev: 916

StdP: 90.80

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94037

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.6	-21.6	-28.9	0.3	-24.3	-25.5	0.4	-21.2	18.8	-0.8	17.5	-3.1	5.1	300	0.736

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	35.3	18.7	33.1	18.3	31.2	18.1	21.4	29.8	20.4	29.0	19.4	27.9	5.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	15.3	24.8	17.7	14.2	23.7	16.7	13.3	23.0	66.5	29.9	62.8	28.7	59.3	27.9	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.3	14.4	12.3	-28.9	38.9	2.4	2.5	-30.6	40.7	-32.0	42.1	-33.3	43.5	-35.0	45.3		
			-29.1	23.0	2.3	1.1	-30.7	23.8	-32.1	24.4	-33.4	25.0	-35.1	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-6.1	-5.8	0.8	6.4	12.0	18.0	22.8	21.3	16.1	7.9	1.1	-4.7
	DBStd	11.78	7.93	8.18	7.84	5.52	4.86	4.03	3.52	3.77	5.08	5.41	6.52	7.47
	HDD10.0	2240	499	444	294	130	34	2	0	0	8	104	270	454
	HDD18.3	4266	757	677	545	357	203	52	6	16	101	324	516	713
	CDD10.0	1352	0	1	7	23	97	243	396	349	192	39	4	0
	CDD18.3	336	0	0	0	1	6	43	143	107	34	2	0	0
	CDH23.3	4367	0	0	4	25	105	570	1734	1337	543	48	0	0
	CDH26.7	1949	0	0	0	4	22	203	853	627	227	12	0	0
Wind	WSAvg	5.5	5.7	5.4	5.9	6.1	6.1	5.5	4.9	4.9	5.2	5.7	5.5	5.5
Precipitation	PrecAvg	384	9	9	17	39	65	75	50	37	27	32	12	8
	PrecMax	517	41	29	54	89	162	136	146	92	77	115	87	26
	PrecMin	203	0	0	2	6	12	32	8	6	2	2	0	0
	PrecStd	84	9	6	13	19	41	28	31	22	19	28	16	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	17.0	23.0	26.4	29.2	35.0	39.0	37.5	34.9	28.1	20.6	13.8
		MCWB	5.6	7.6	10.2	12.4	16.3	19.2	19.4	19.4	17.7	14.2	9.7	5.3
	2%	DB	8.5	11.3	18.7	22.3	26.3	31.3	36.2	34.8	32.1	23.8	16.0	10.0
		MCWB	3.6	4.9	8.6	11.1	15.6	18.5	19.3	18.1	16.8	12.4	7.6	3.7
	5%	DB	6.0	7.2	15.3	19.0	23.8	28.8	33.6	32.6	29.3	20.1	12.8	7.3
		MCWB	2.1	2.9	6.9	9.6	14.1	18.1	18.9	18.1	15.5	10.5	5.9	2.3
	10%	DB	3.9	4.5	11.8	16.2	21.3	26.7	31.5	30.4	26.4	17.0	10.1	4.6
		MCWB	0.9	1.3	5.6	8.2	13.2	17.0	18.8	17.8	14.7	9.4	4.6	1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	7.8	11.1	13.4	18.8	22.2	23.0	21.9	19.9	15.1	10.1	6.3
		MCDB	11.8	15.8	21.0	24.2	26.4	28.7	29.8	31.2	29.4	25.4	19.6	12.2
	2%	WB	3.8	5.1	9.1	11.7	16.7	20.6	21.8	20.6	18.0	13.2	8.0	4.2
		MCDB	8.0	10.6	17.4	21.0	23.3	27.6	30.2	29.8	28.2	21.5	15.2	8.8
	5%	WB	2.3	3.1	7.5	10.2	15.1	19.2	20.8	19.6	16.8	11.4	6.3	2.6
		MCDB	5.6	6.9	15.0	18.1	22.3	26.5	29.6	28.7	26.8	18.7	12.5	6.9
	10%	WB	1.0	1.4	5.8	8.8	13.7	18.0	19.9	18.7	15.4	9.8	4.7	1.1
		MCDB	3.8	4.3	11.5	15.5	20.3	25.4	28.5	27.4	24.3	16.3	9.5	4.6
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.6	11.9	12.5	12.0	13.0	14.7	14.9	14.5	12.3	11.5	9.7
		MCDBR	11.8	13.7	17.6	17.9	16.0	16.6	18.1	18.2	19.0	17.8	15.4	11.9
		MCWBR	8.0	8.4	9.3	9.1	7.6	7.2	5.9	6.5	7.1	8.6	8.6	7.6
	5% WB	MCDBR	11.6	13.0	16.7	16.9	14.6	15.0	15.4	15.3	17.0	16.5	15.1	11.8
		MCWBR	8.1	8.3	9.2	8.9	7.7	7.3	6.3	6.4	6.8	8.2	8.7	8.0
Clear-Sky Solar Irradiance	taub	0.252	0.257	0.276	0.311	0.330	0.344	0.360	0.359	0.324	0.289	0.261	0.246	
	taud	2.546	2.547	2.551	2.443	2.461	2.452	2.434	2.412	2.500	2.566	2.543	2.546	
	Ebn at Noon	888	948	969	950	937	921	901	889	899	894	866	859	
	Edh at Noon	62	75	86	106	108	109	110	108	91	73	61	55	
All-Sky Solar Radiation	RadAvg	1.79	2.79	3.91	5.00	5.89	6.77	7.04	5.93	4.55	2.95	1.94	1.45	
	RadStd	0.11	0.20	0.31	0.29	0.59	0.50	0.29	0.32	0.29	0.29	0.18	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page